Evidence Appendix

Air pollution near the ground, why height matters, and the design of our CanSat

Team Ironclad | Oakridge International School, Gachibowli, Hyderabad

SEVA FIRST Innovation Challenge 2026 | National level | NAT-004-PS, Autonomous CanSat

Sishir Kotra (leader), Dhanush Musunuru | Mentor: Agin M

What this appendix is

- Every number here comes from a published source or from **real government monitoring data** that we downloaded and analysed ourselves. Each chart names its source underneath.
- We analysed **2 years of 15-minute readings (2024-25) from all 14 Hyderabad-area stations**: about 980,000 time slots in total. Method and caveats are in Part G.
- The 10 papers and reports we rely on are included as PDFs in the **papers** folder of this archive, with a one-page summary of each in Part F.
- Design numbers marked 'from our CAD' or 'calculated' are ours; nothing here is a test result yet. Test results will go into the Feedback and Test Log spreadsheet.

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At a glance: 16 numbers

India

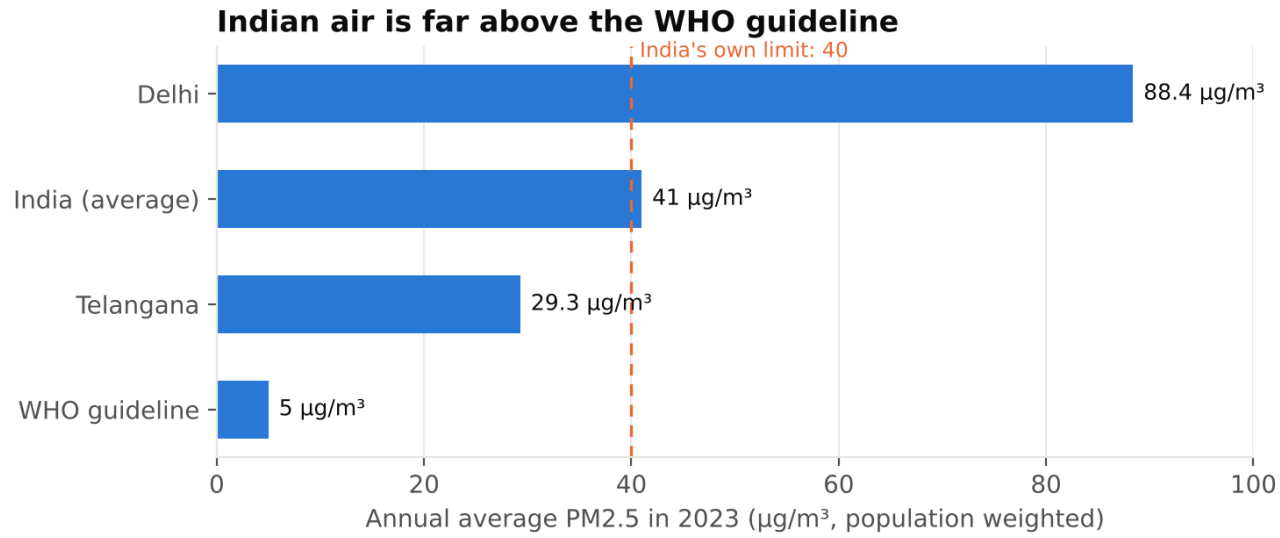
41 µg/m³ India's average PM2.5 in 2023, 8x the WHO guideline (AQLI)	**1.4 billion** Indians living above the WHO guideline: everyone (AQLI)	**3.5 years** of life the average Indian loses to particulate pollution (AQLI)	**1.67 million** deaths from air pollution in India in 2019, 17.8% of all (GBD)
$36.8 bn lost output in 2019, 1.36% of India's GDP (GBD)	**28 of 130** clean-air programme cities with no continuous station (CREA)	**465** manual stations still missing against the 2024 target (CREA)	**103 of 231** cities above India's PM2.5 limit in 2025 (CREA)

Hyderabad: our analysis of 2024-25 station data

587 of 730 days above WHO's daily PM2.5 guideline (6-station median)	**98** days above India's own daily limit of 60 (6-station median)	**135** days over the daily limit at Zoo Park in 2025 alone	**52.6 µg/m³** Zoo Park's 2025 average, 10x the WHO guideline
34-63% more PM2.5 on winter nights than in the afternoon	**8 of 14** stations that barely change between monsoon and winter	**6 of 14** stations missing over 20% of their readings	**+60%** PM2.5 at 100 m vs ground during Delhi haze (Ahlawat 2026)

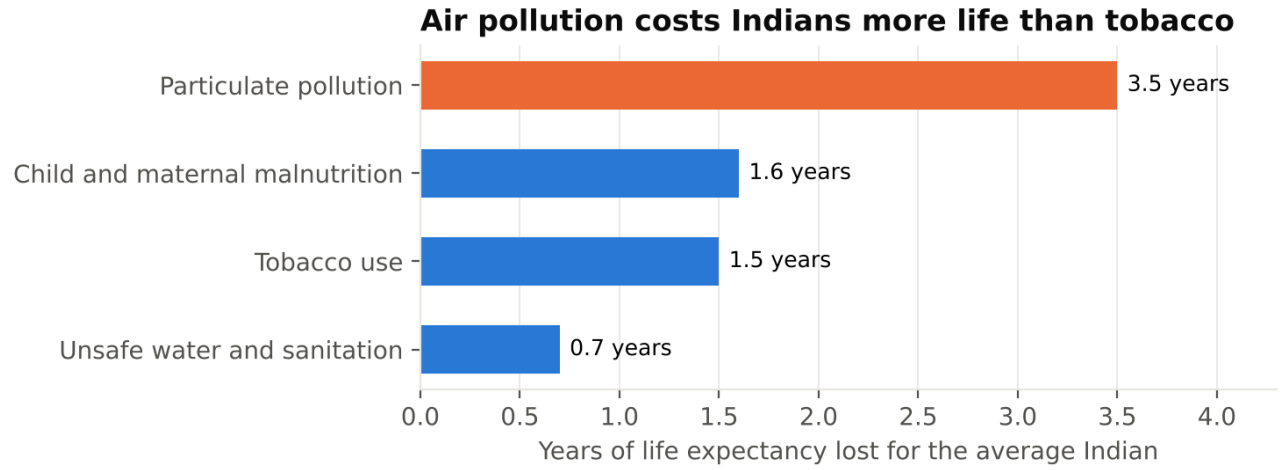
What it adds up to: pollution is everywhere, it changes with the hour and the height, and the ground network that measures it has gaps. A cheap, reusable CanSat that measures PM2.5 from 500 m down to the ground fills one of those gaps.

Part A. The problem across India



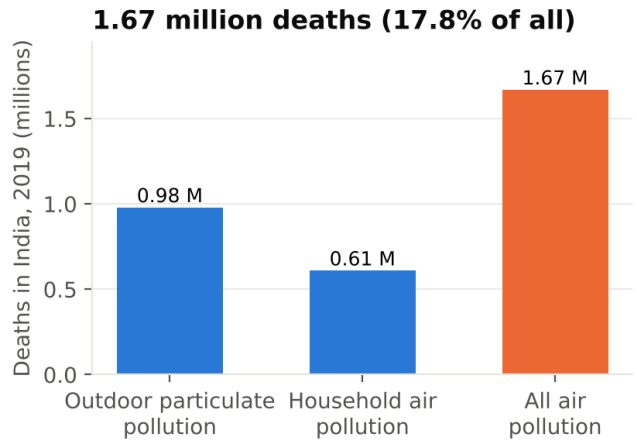
Source: AQLI India Fact Sheet 2025 (EPIC, Univ. of Chicago), data year 2023

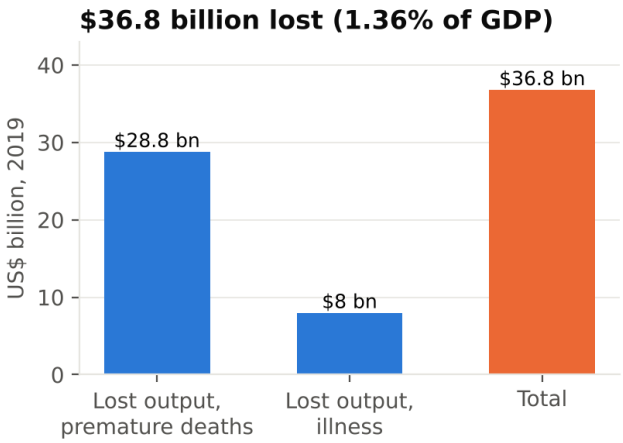
- India's population-weighted PM2.5 in 2023 was **41 µg/m³**, just above India's own limit of 40 and over 8 times the WHO guideline of 5.
- Telangana averaged **29.3 µg/m³**: under India's limit, but nearly 6 times WHO's. AQLI says Telangana residents lose about **2.4 years** of life to it.
- 46% of Indians live where the national limit is broken; all 1.4 billion live above the WHO guideline.



Source: AQLI India Fact Sheet 2025; unsafe water = 8.4 months

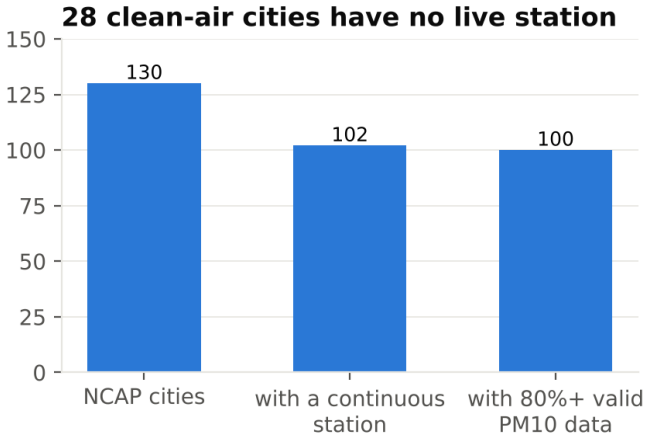
- Particulate pollution takes more life expectancy (3.5 years) than child and maternal malnutrition (1.6) or tobacco (1.5).
- Delhi residents would gain 8.2 years if air met the WHO guideline (AQLI).

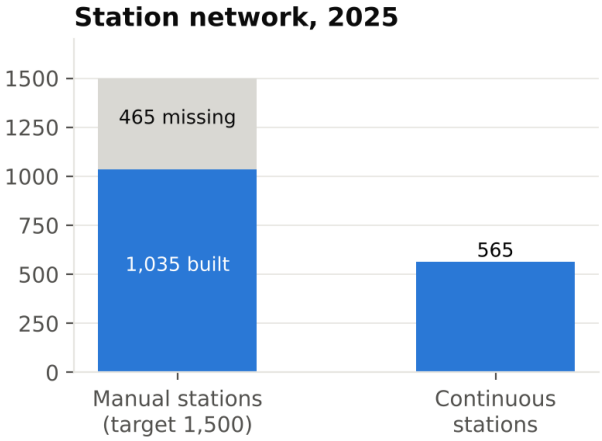




Source: India State-Level Disease Burden Initiative, Lancet Planetary Health 2021; 5: e25-38 (GBD 2019)

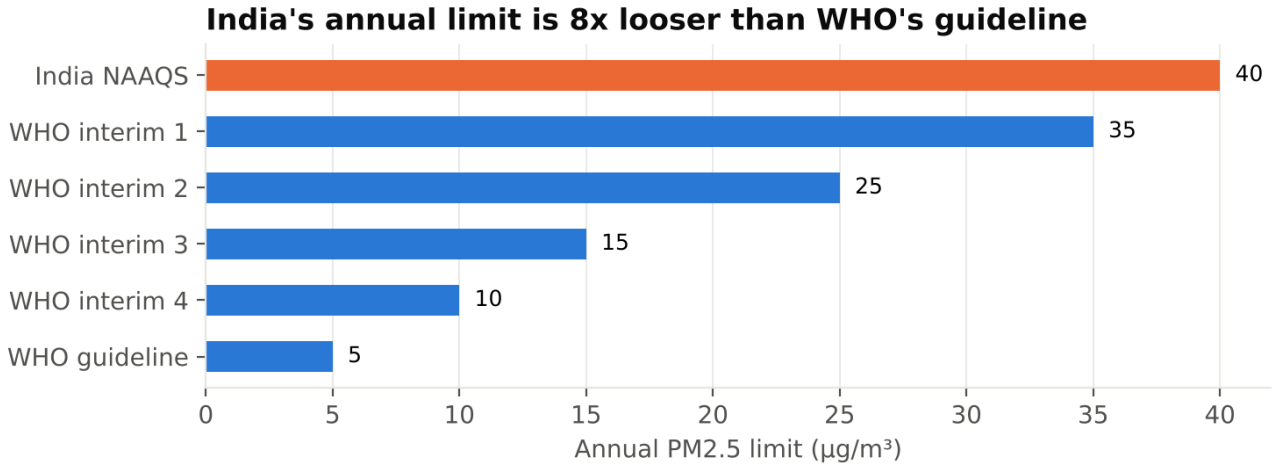
- GBD 2019: **1.67 million** deaths (95% range 1.42-1.92 million) were linked to air pollution in 2019, 17.8% of all deaths in India.
- Death rate from outdoor particulate pollution rose **115%** between 1990 and 2019, while household air pollution fell 64%.
- Telangana lost about **US$1.1 billion** in output in 2019 (US$28.7 per person), from the same study's state table.





Source: CREA, Tracing the Hazy Air 2026 (NCAP progress report)

- In 2025 India had 1,600 stations: 565 continuous and 1,035 manual, covering 584 cities.
- 28 of 130 cities in the National Clean Air Programme still have no continuous station, and the manual-station target of 1,500 is 465 short.
- CREA notes that even where stations exist, **data quality remains a concern** due to poor maintenance and siting. Our Hyderabad analysis (Part B) finds the same.



Source: WHO Global Air Quality Guidelines 2021 (Table 0.1); CPCB NAAQS 2009

- India's annual PM2.5 standard (40 µg/m³, set in 2009) is looser than every WHO interim target except the first. A city can 'pass' in India and still be 8 times over WHO's guideline.
- Daily: India allows 60 µg/m³ in 24 hours; WHO recommends 15.

What people say: published surveys

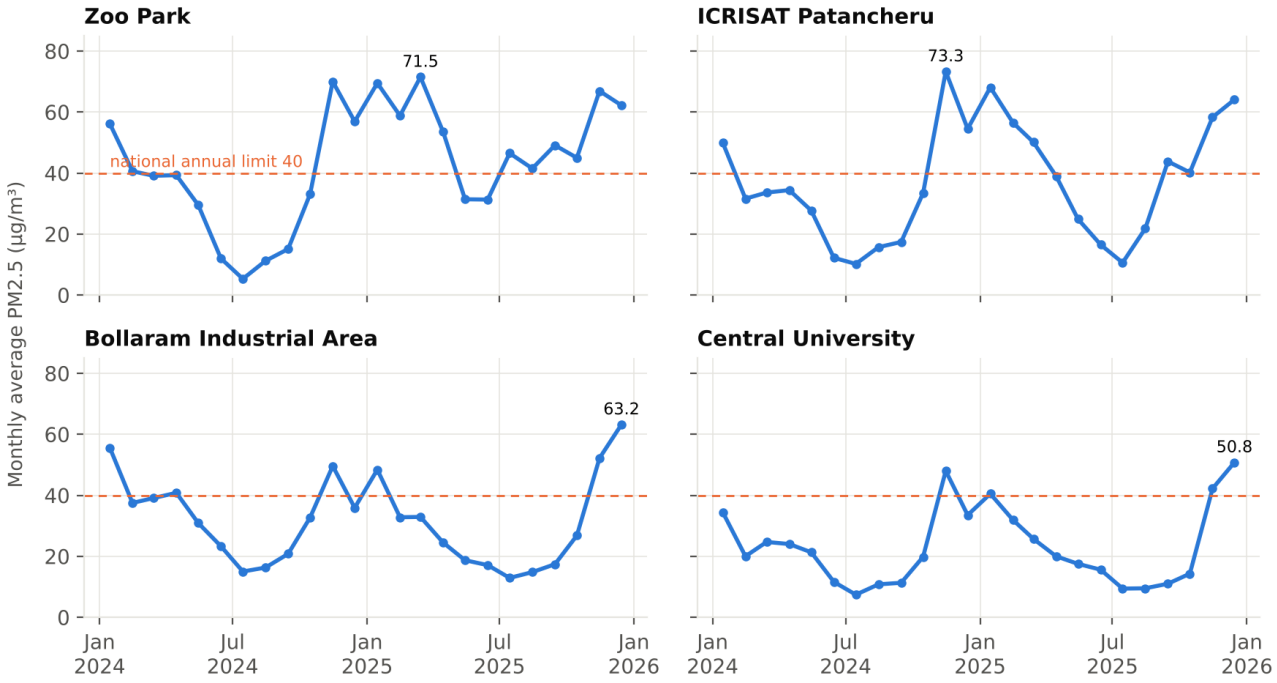
We did not run these surveys. They are published results used as evidence that people feel this problem.

Survey	Who answered	Key findings
LocalCircles, released 15 Dec 2025	34,000+ responses, Delhi-NCR	82% know someone close with a serious pollution-linked illness (28% know four or more). 73% are very worried about affording future healthcare. 8% plan to leave the NCR.
Yale Program on Climate Change Communication with CVoter, Dec 2024 to Feb 2025	10,751 adults, nationally representative	59% of city residents were very worried severe air pollution could harm their area, against 52% in rural areas.

Part B. Hyderabad: what two years of station data show

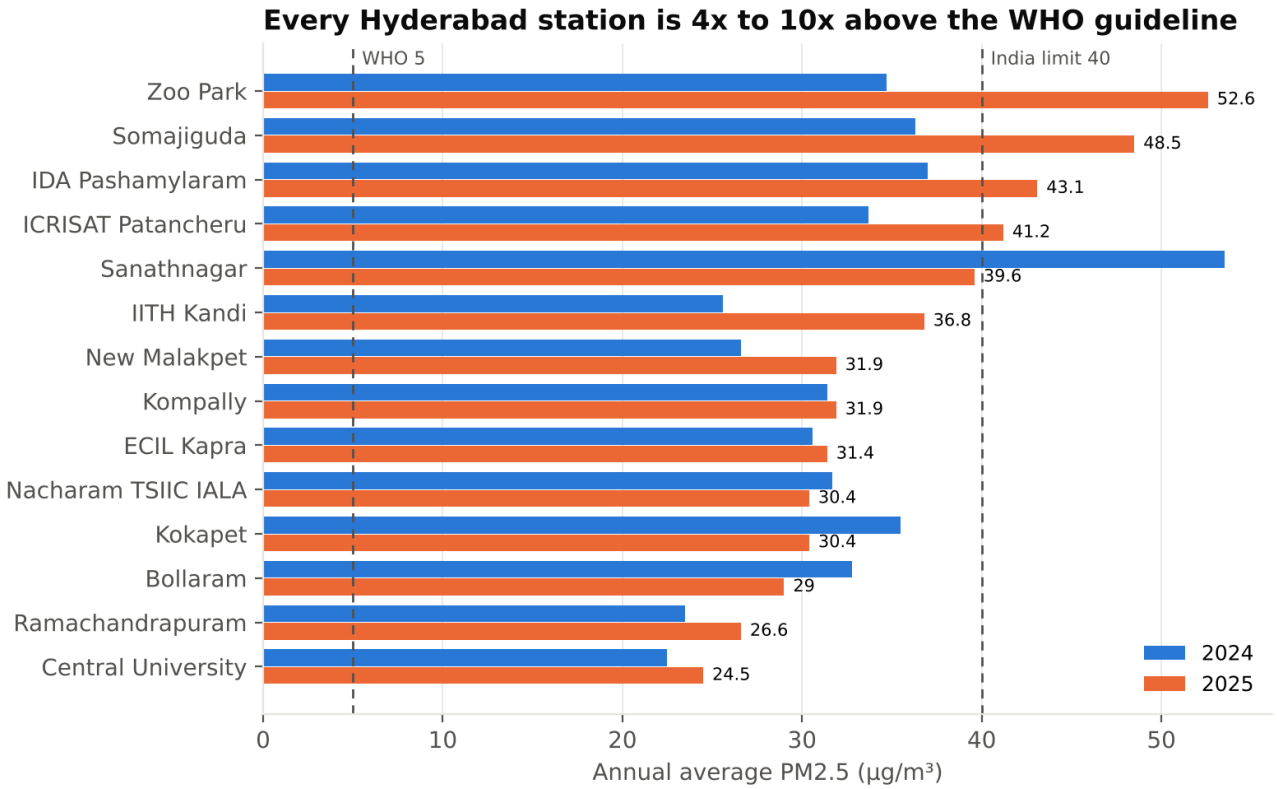
We downloaded every 15-minute reading from 1 Jan 2024 to 31 Dec 2025 for all 14 Hyderabad-area continuous stations (CPCB and Telangana Pollution Control Board data, published by OpenCity). We averaged each day that had at least 48 of 96 valid readings, then built monthly, yearly and hour-of-day averages. Full method in Part G; the output tables are in the **data** folder.

Hyderabad: pollution climbs every winter (Nov to Feb) and falls in the monsoon



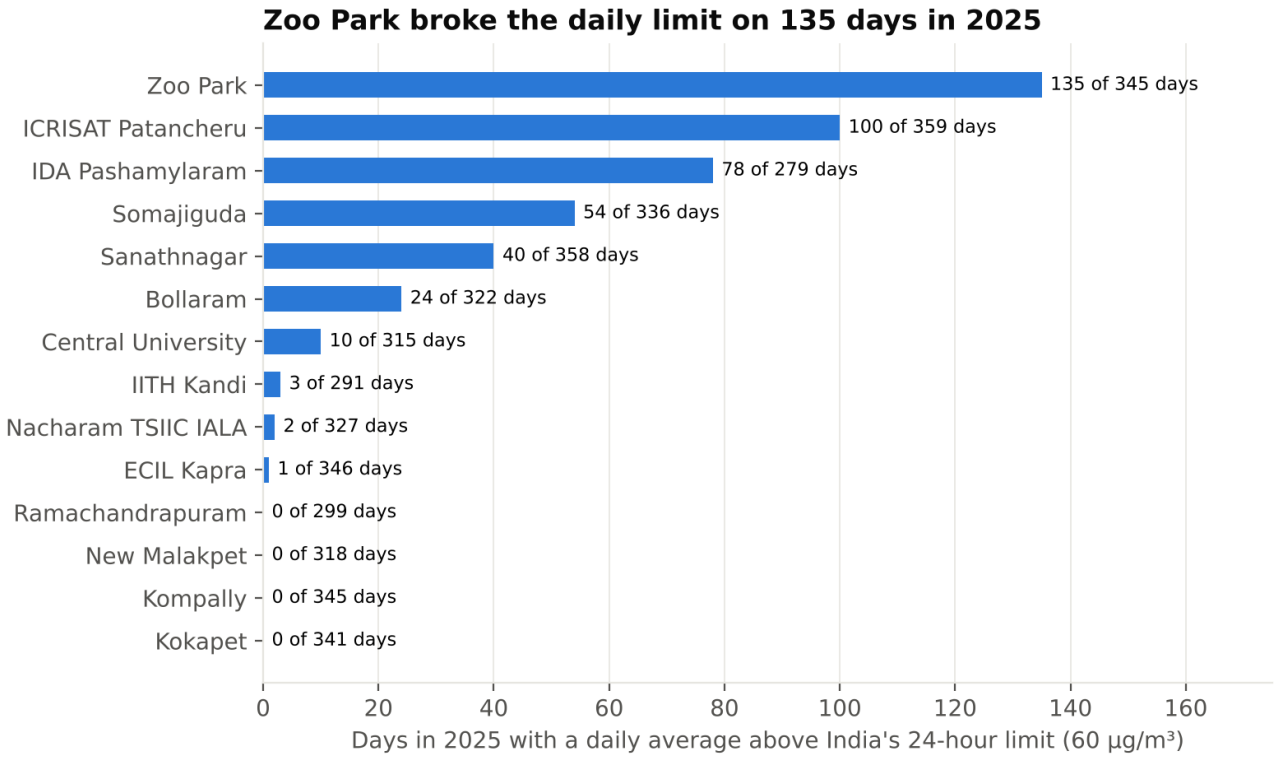
Source: CPCB/TSPCB 15-minute station data 2024-25, via OpenCity (data.opencity.in); our averages, days with 48+ valid readings

- All four stations with a clean record follow the same cycle: lowest in July and August (monsoon rain washes particles out), highest from November to February.
- Winter monthly averages reached **73.3 µg/m³** (ICRISAT, Nov 2024) and **71.5** (Zoo Park, Mar 2025), well above the annual limit of 40.
- Central University, the station closest to our school in Gachibowli, still reached **48 µg/m³** in Nov 2024 and **51** in Dec 2025.



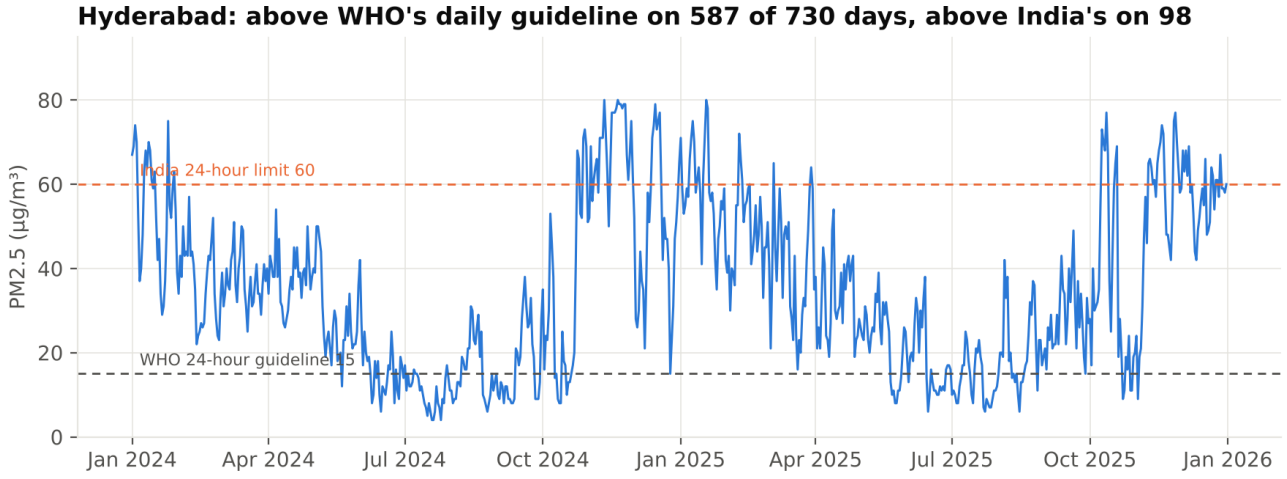
Source: CPCB/TSPCB station data via OpenCity; mean of valid daily averages

- No station met the WHO guideline in either year. The cleanest (Central University, 22.5 and 24.5) was about 5 times over it.
- Zoo Park rose from 34.7 to 52.6 µg/m³ between 2024 and 2025; Somajiguda from 36.3 to 48.5.



Source: CPCB/TSPCB station data via OpenCity; days with 48+ valid 15-minute readings

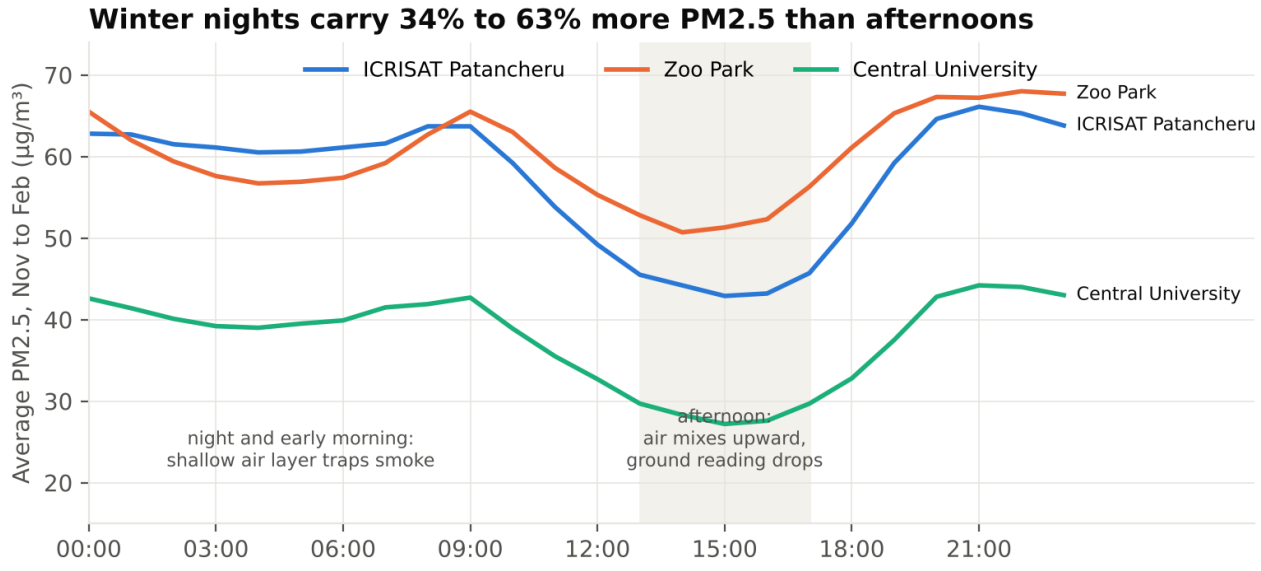
- Zoo Park, ICRISAT and IDA Pashamylaram each broke India's daily limit on 78 to 135 days in 2025.
- Seven stations reported almost no days over the limit; several of these are the same stations flagged for flat readings in chart B5.



Source: CPCB/TSPCB data via OpenCity; daily median of the 6 stations with a normal seasonal pattern (3+ reporting)

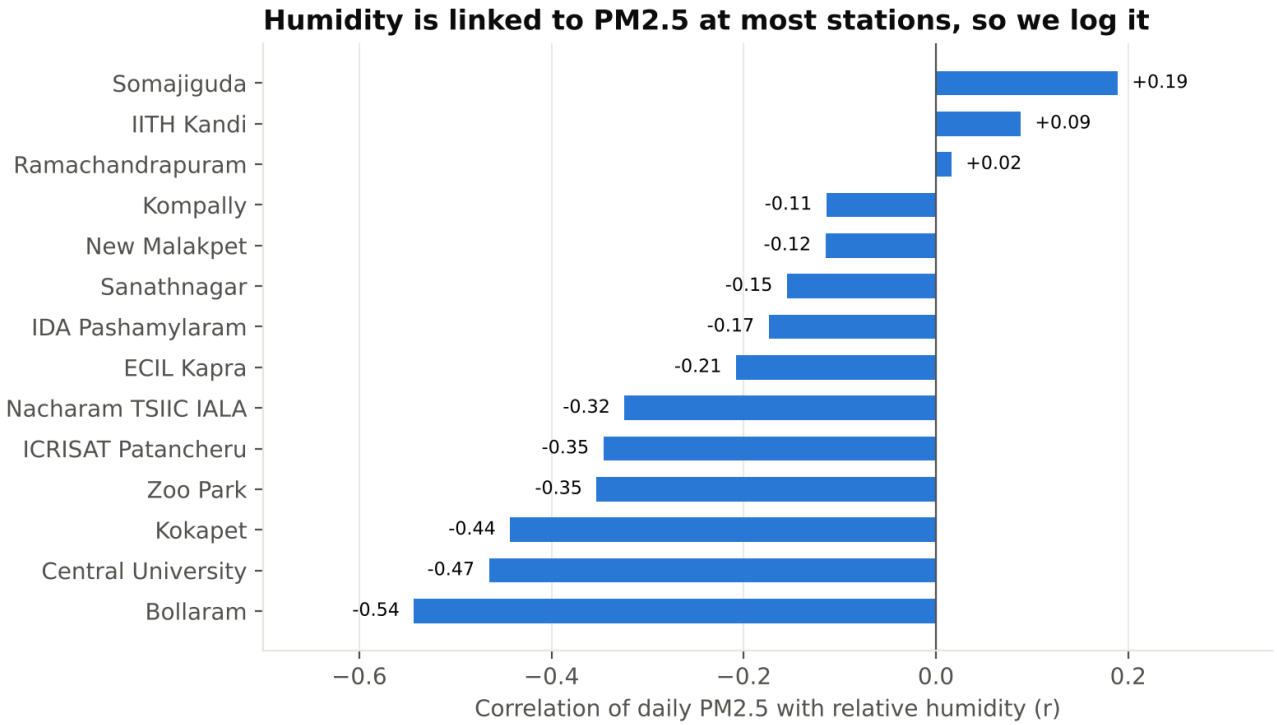
- Using the median of the 6 stations with a normal seasonal pattern, Hyderabad was above WHO's daily guideline (15 µg/m³) on **587 of 730** days with data and above India's limit (60) on **98**.
- Clean days cluster in the monsoon; from October the line climbs fast.

The daily cycle: why the time and height of a reading matter



Source: CPCB/TSPCB station data via OpenCity, Nov-Feb of 2024 and 2025; hour-of-day averages (station local time)

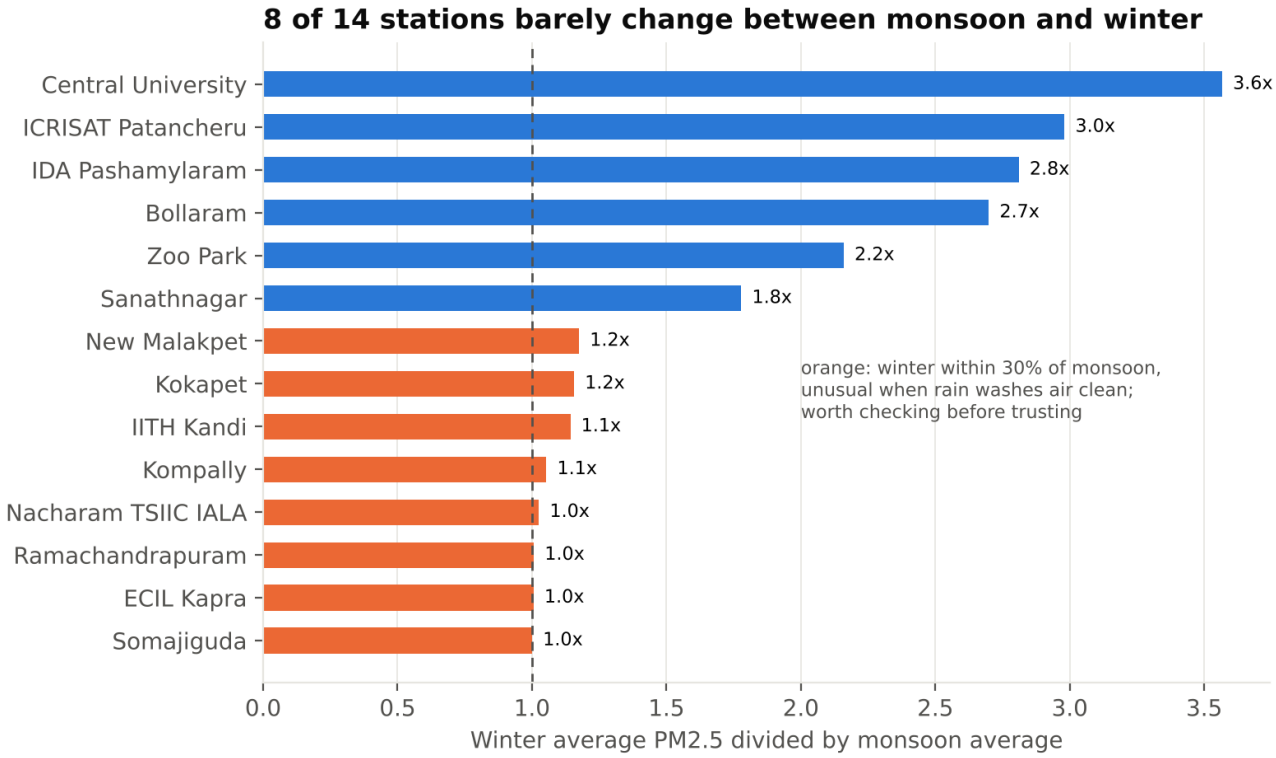
- On winter days, ground readings peak at night and in the morning and fall by a third or more in the afternoon. The night peak is **34% to 63%** above the afternoon low at these three stations.
- The usual explanation is the mixing layer: in the afternoon, the sun heats the ground and pollution mixes upward; at night a shallow layer traps it near the ground. A ground station sees only the bottom of this layer. A CanSat falling from 500 m sees the whole column.
- In Delhi, a drone study found PM2.5 **60% higher at 100 m** than at ground level during haze (Part C).



Source: Our analysis of CPCB/TSPCB station data via OpenCity, 2024-25 daily averages

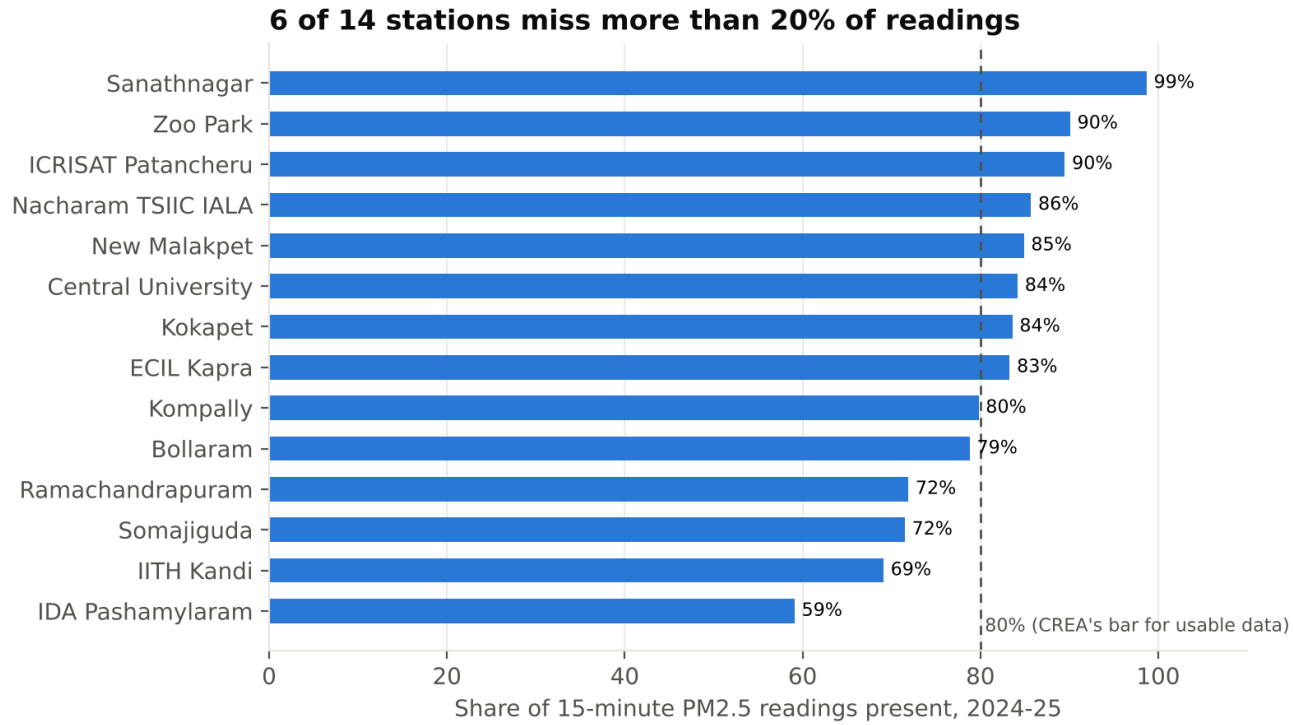
- Daily PM2.5 and relative humidity are linked at most stations (negative here because the humid monsoon is also the clean season).
- Optical sensors like ours also **over-read in humid air**, because particles swell with water. The US EPA correction for this sensor type uses humidity directly (Part C). So we log humidity with every reading.

How reliable is the ground network?



Source: Our analysis of CPCB/TSPCB station data via OpenCity; winter = Nov-Feb, monsoon = Jun-Sep

- At 6 stations, winter PM2.5 is 1.8 to 3.6 times the monsoon level, as expected.
- At **8 stations**, winter and monsoon averages are within 30% of each other, and their hour-by-hour curves are nearly flat. This is unusual for a city where rain clears the air every July. We cannot say why from the data alone (possible causes include calibration drift or stuck values), but it means **these readings should be checked before they are trusted**.
- This matches CREA's national finding that station data quality is a concern.



Source: Our analysis of CPCB/TSPCB station data via OpenCity

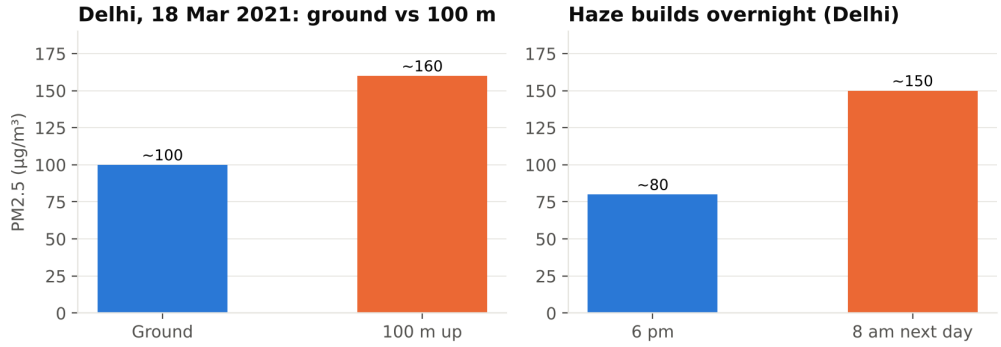
- 6 of 14 stations are missing over 20% of their 15-minute PM2.5 readings across the two years.
- Sanathnagar recorded a daily average of **887 µg/m³** on 28 Apr 2024, far above any other station that day: almost certainly a faulty reading. We excluded no data, but we used medians in chart B6 to limit the effect.

All 14 stations at a glance

Station	2024 avg	2025 avg	Days >60 (2025)	Data present	Winter ÷ monsoon	Note
Zoo Park	34.7	52.6	135 of 345	90%	2.2x	-
Somajiguda	36.3	48.5	54 of 336	72%	1.0x	flat season pattern; gaps
IDA Pashamylaram	37	43.1	78 of 279	59%	2.8x	gaps
ICRISAT Patancheru	33.7	41.2	100 of 359	90%	3.0x	-
Sanathnagar	53.5	39.6	40 of 358	99%	1.8x	spike 887 on 2024-04-28
IITH Kandi	25.6	36.8	3 of 291	69%	1.1x	flat season pattern; gaps
Kompally Municipal Office	31.4	31.9	0 of 345	80%	1.1x	flat season pattern; gaps
New Malakpet	26.6	31.9	0 of 318	85%	1.2x	flat season pattern
ECIL Kapra	30.6	31.4	1 of 346	83%	1.0x	flat season pattern
Kokapet	35.5	30.4	0 of 341	84%	1.2x	flat season pattern
Nacharam TSIIC IALA	31.7	30.4	2 of 327	86%	1.0x	flat season pattern
Bollaram Industrial Area	32.8	29	24 of 322	79%	2.7x	gaps
Ramachandrapuram	23.5	26.6	0 of 299	72%	1.0x	flat season pattern; gaps
Central University	22.5	24.5	10 of 315	84%	3.6x	-

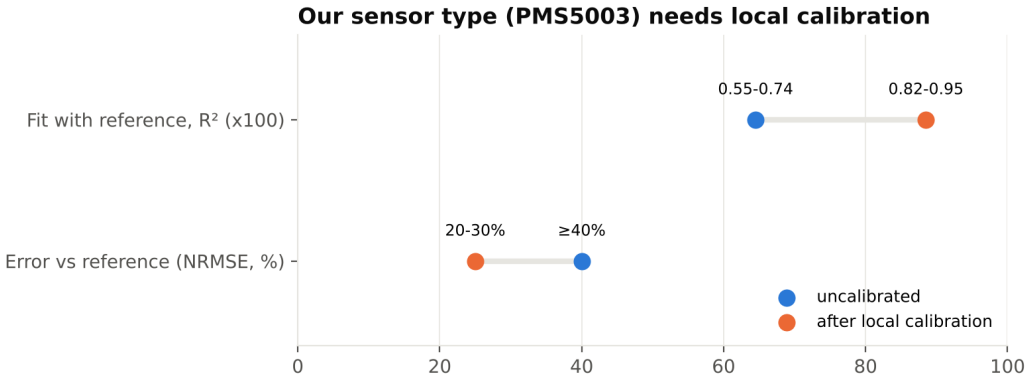
Averages in µg/m³. 'Days >60' counts days with a valid daily average. Source: our analysis of CPCB/TSPCB data via OpenCity (data/hyderabad_station_summary_2024_25.csv).

Part C. Why height matters, and can a cheap sensor be trusted?



Source: Ahlawat et al., npj Clean Air 2 (2026) 8, doi:10.1038/s44407-025-00044-3; values approximate, as reported

- Ahlawat et al. (2026) flew a drone carrying a low-cost PM sensor over Delhi in March 2021. During haze, PM2.5 was about **160 µg/m³ at 100 m**, 60% higher than the roughly 100 µg/m³ at ground level.
- Haze built overnight: PM2.5 rose from about 80 µg/m³ at 6 pm to about 150 µg/m³ at 8 am, while humidity stayed above 80%.
- A leading air quality model underestimated morning haze PM2.5 by **52.6%**, because it got humidity wrong. The authors conclude drone-mounted low-cost sensors are a valuable tool for vertical air quality data.



Source: Campmier et al., Atmos. Meas. Tech. 16, 4357-4374 (2023); PurpleAir PA-II (2x PMS5003), 3 Indian sites; range midpoints

- Campmier et al. (2023) ran PurpleAir monitors, which contain two Plantower PMS5003 sensors (the same sensor we use), next to government reference monitors in Delhi, rural Hamirpur and Bengaluru for over a year.
- Raw readings tracked the reference (R^2 0.55 to 0.74) but were off by 40% or more. After local calibration, error fell to **20 to 30%** and R^2 rose to **0.82 to 0.95**. A calibration made in one season did not transfer well to another.
- **What we do with this:** before flying, we place our sensor beside a CPCB station (Central University is nearest to us) for at least two weeks, fit a simple correction, and publish its error. That is our 'readings you can trust' claim.

The humidity correction we will start from

The US EPA's nationwide correction for PurpleAir (PMS5003) data, from about 12,000 days of side-by-side readings in 16 US states (Barkjohn et al., 2021):

$$\text{PM2.5 (corrected)} = 0.524 \times \text{PM2.5 (sensor, CF=1)} - 0.0862 \times \text{RH} + 5.75$$

- It cut the error from 8 to 3 µg/m³ in the US. Indian aerosols differ, so we will refit the three numbers against the Hyderabad CPCB station, as Campmier et al. recommend.
- This is why the BME280 humidity reading is part of every data line.

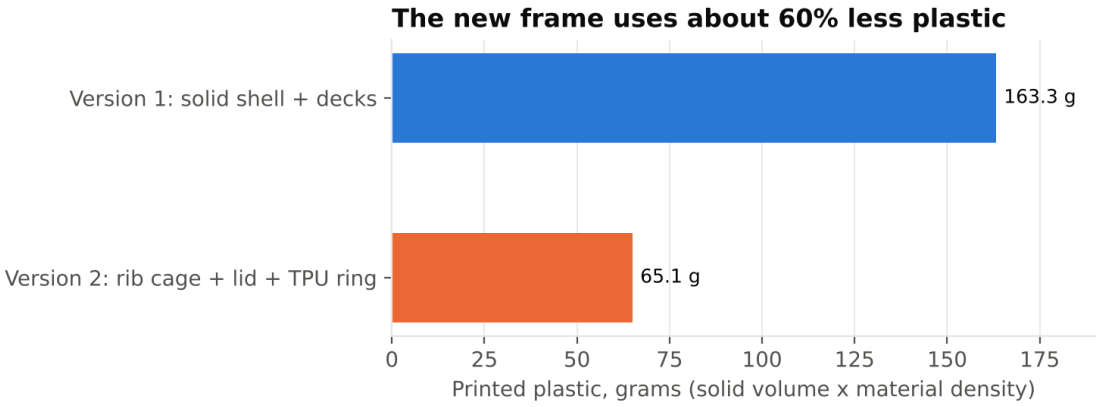
Part D. What others have built, and what is new in ours

Project	What it measured	How	Gap we address
NAMBI-VJ CanSat, Vidya Jyothi Institute of Technology, Hyderabad (arXiv 2024)	Gas levels (PPM, CO2), GPS; dropped from 900 m by drone	310 x 125 mm, about 1 kg; Zigbee radio; spill-hole parachute at 3 m/s	No PM2.5; no check of sensor accuracy against a station; too heavy and too big for the 700 g limit
Onboard computer for CanSat atmosphere monitoring, MIT World Peace University, Pune (arXiv 2023)	Gases, UV, temperature, pressure, GPS, motion	ESP32 and a radio transceiver	Design of the electronics only; no flight PM2.5 profile or sensor validation
Vertical CanSatAIr, Liceo P. Galluppi, Tropea, Italy (Global Schools Prize 2026 finalist)	PM2.5, PM10, VOCs, temperature, pressure, humidity	SPS30 and BME680 sensors, LoRa to a Raspberry Pi; drone test at about 120 m	Closest to ours. We add a published error against an official station, a live PM2.5-vs-height chart, a two-way 'find me' command and an Indian city context
Drone haze study, Delhi (Ahlawat et al., npj Clean Air 2026)	PM1, PM2.5 profiles to about 100 m	Research drone with an optimised low-cost sensor inlet	Needs a drone and pilot for every profile. A CanSat costs about ₹10,700 in parts and gives a full 500 m profile per drop
IMD radiosondes (weather balloons)	Temperature, humidity, wind; not pollution	Single-use, about 36,000 a year (2016)	No PM2.5; not reused

Our CanSat in one table

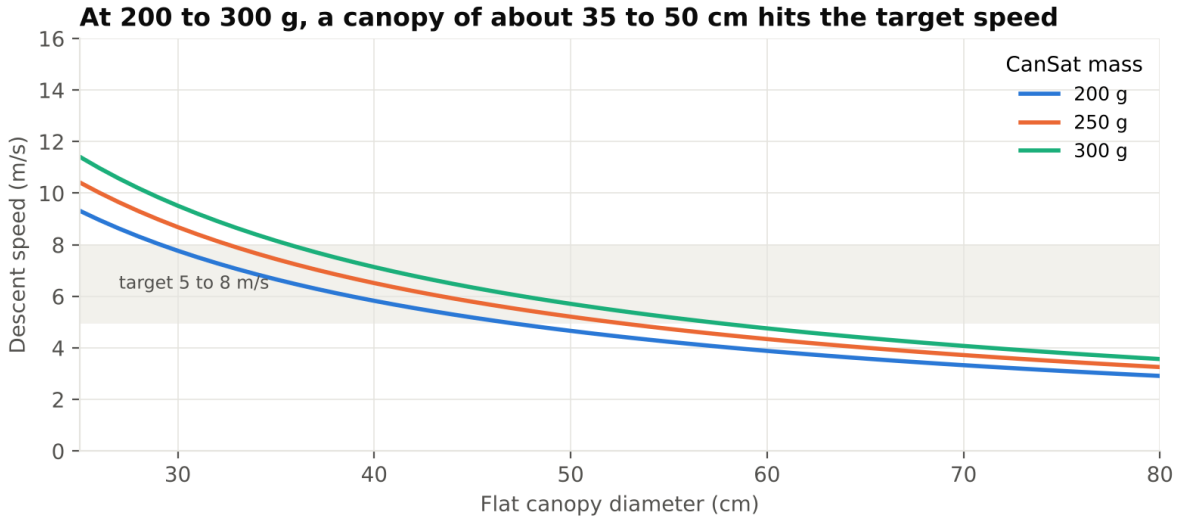
Feature	What the rules ask	What we add
Air quality	A working air quality sensor	PMS5003 checked side by side with a CPCB station; error published
Live data	At least once a second	Live PM2.5-against-height chart as it falls
Humidity	Not required	Logged with every reading, used in the correction
Radio	Downlink; uplink optional	Two-way: laptop can switch on a 'find me' buzzer after landing
Structure	700 g or less; survives landing	Open rib-cage frame (about 60% less plastic) and a flexible TPU crush ring
Parachute release	Descent control	Burn wire frees a spring-loaded lid; resets with a new piece of thread
After the event	Post-flight report	Open design, code and data for about 10,000 school tinkering labs

Part E. Design evidence



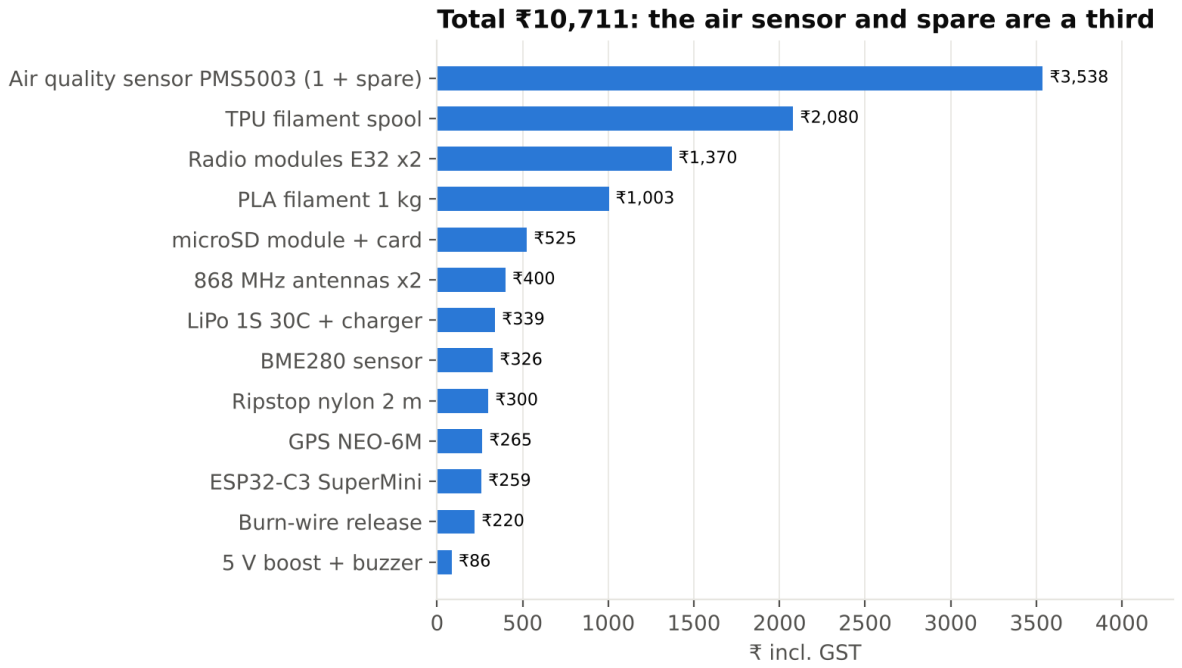
Source: Our CAD models (cadquery volumes); PLA 1.24 g/cm³, TPU 1.21 g/cm³; real prints with infill will be lighter

- From our CAD: the version 2 frame, lid and crush ring together are 65 g if printed solid, against 163 g for the version 1 shell and decks. Real prints use infill, so both will be lighter.
- All parts fit inside the 66 x 115 mm can with no overlaps (checked in the CAD with each part at its datasheet size; wires not modelled).



Source: ESA Teach with Space T10, equation 3 (mg = ½·Cd·ρ·A·v²); Cd 0.8 for a flat canopy, air density 1.15 kg/m³ (Hyderabad, ~500 m)

- The ESA CanSat guide gives the balance at terminal speed: weight = ½ × drag coefficient × air density × canopy area × speed².
- We expect the finished CanSat to weigh roughly 200 to 300 g. For 5 to 8 m/s, that points to a flat canopy of about **35 to 50 cm**. We will weigh the real build and then cut the cloth, and check the speed in drop tests.



Source: Supplier prices checked 11-12 Oct 2026 (Robu, Probots, Hubtronics and others; see slide 12)

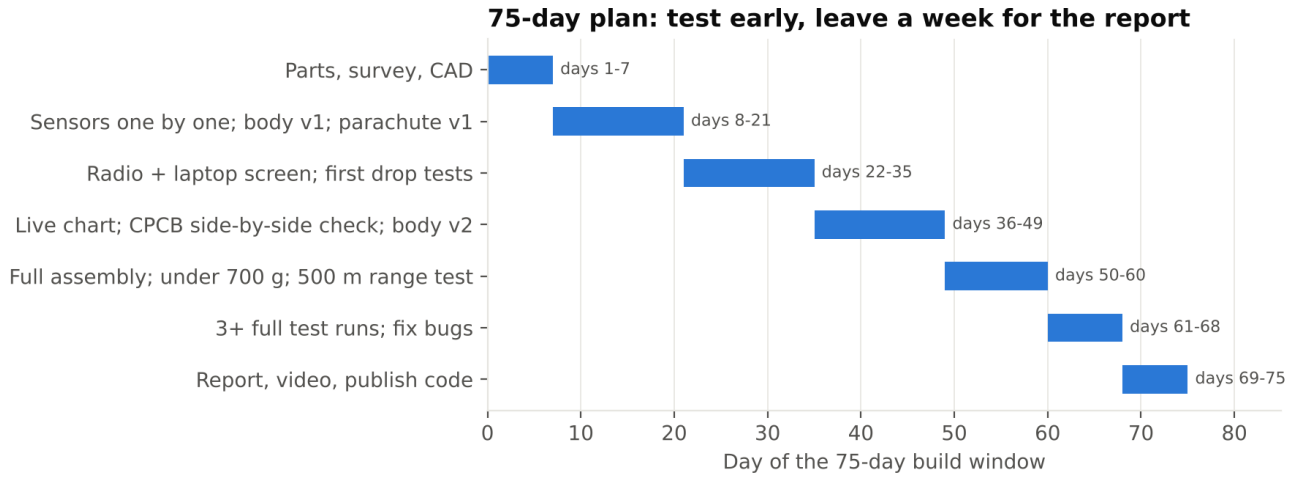
- Total ₹10,711 including a spare air sensor and a TPU spool. Without the TPU spool (if printed with the mentor's lab filament): ₹8,631.
- Item prices and suppliers are on slide 12 of the main presentation.

Power design (fixed in version 2)

Issue found in review	Why it matters	Fix
PMS5003 fan needs 4.5 to 5.5 V	A 1-cell LiPo gives 3.7 V; the sensor would not run reliably	MT3608 boost board makes 5 V for the sensor and the ESP32-C3's 5 V pin
AMS1117 regulator needs about 4.4 V in for 3.3 V out	Would brown out on a LiPo	Removed; the ESP32-C3 SuperMini's own regulator supplies 3.3 V
Burn wire draws 1 to 2 A for about a second	An ordinary small LiPo could sag and restart the ESP32 mid-fall	1S 360 mAh 30C drone-type LiPo, rated for high current

How we meet the problem statement

Requirement (NAT-004-PS)	Our approach	How we will prove it
Total mass 700 g or less	Rib-cage frame, ESP32-C3, small LiPo	Weigh after every change
Released from a drone at 500 m	Works on its own after release; no control needed	Drop tests under mentor supervision, below 120 m (Drone Rules 2021)
Descent 5 to 8 m/s	Parachute sized from the ESA formula after weighing	Speed from the pressure log in each drop
Live telemetry at least once a second	E32 radio at 865 to 867 MHz (licence-free)	Count lost lines in range tests up to 600 m
Altitude, GPS, battery and mission data	BME280, NEO-6M, battery voltage, PMS5003	Check every field in the log
Backup copy on board	microSD card	Compare the card and the laptop log line by line
Safe recovery	TPU crush ring, buzzer, bright parachute	Repeat drops; inspect for damage



Source: Team Ironclad plan (slide 14)

- Every phase ends with a test that has a pass mark (see slide 14). A spare week before submission covers exams and delays.

Part F. The papers and reports behind this appendix

All ten are included as PDFs in the papers folder, named by the number below. Summaries are in our own words; key numbers are as reported by the authors.

01. Drone measurements reveal high near-surface urban haze

Ahlawat A. et al. (2026). npj Clean Air 2, 8. doi:10.1038/s44407-025-00044-3. Open access (CC BY).

What they did: Flew a drone with an optimised low-cost PM sensor over South Delhi in March 2021 to measure PM2.5 by height during post-winter haze, and compared it with the WRF-Chem air quality model.

Key findings:

- PM2.5 about 160 µg/m³ at 100 m, 60% higher than at ground level.
- Under humid conditions (RH above 70%), the PM1/PM2.5 ratio points to haze driven by particles that absorb water.
- The model underestimated morning haze PM2.5 by 52.6 ± 5.5%, linked to a 30% dry bias in its humidity.

How we use it: Our main evidence that ground stations miss what happens higher up, and that humidity matters.

02. Seasonally optimized calibrations improve low-cost sensor performance: long-term field evaluation of PurpleAir sensors in urban and rural India

Campmier M. J. et al. (2023). Atmospheric Measurement Techniques 16, 4357-4374. doi:10.5194/amt-16-4357-2023. Open access (CC BY 4.0).

What they did: Placed PurpleAir PA-II monitors (each with two Plantower PMS5003 sensors) beside reference monitors in Delhi, Hamirpur and Bengaluru for long periods.

Key findings:

- Raw data: R^2 0.55 to 0.74 but error (NRMSE) of 40% or more.
- Annual calibration: error 20 to 30%, R^2 0.82 to 0.95.
- Local, season-aware calibration works best; a short calibration does not transfer reliably to other seasons.

How we use it: Shows our exact sensor type works in India once calibrated locally. Basis for our CPCB side-by-side check.

03. Development and application of a United States-wide correction for PM2.5 data collected with the PurpleAir sensor

Barkjohn K. K., Gantt B., Clements A. L. (2021). Atmospheric Measurement Techniques 14, 4617-4637. doi:10.5194/amt-14-4617-2021. Open access.

What they did: Compared almost 12,000 days of PurpleAir data with regulatory monitors across 16 US states.

Key findings:

- Raw PurpleAir data overestimated PM2.5 by about 40%.
- Adding relative humidity to a linear correction reduced bias further.
- Final correction: PM2.5 = 0.524 × PA(cf_1) − 0.0862 × RH + 5.75; error fell from 8 to 3 μg/m³.

How we use it: Our starting correction formula, and the reason humidity is in every data line.

04. Health and economic impact of air pollution in the states of India: the Global Burden of Disease Study 2019

India State-Level Disease Burden Initiative Air Pollution Collaborators (2021). Lancet Planetary Health 5, e25-38. doi:10.1016/S2542-5196(20)30298-9. Open access (CC BY 4.0).

What they did: Estimated deaths, disease and economic loss from air pollution for every Indian state in 2019.

Key findings:

- 1.67 million deaths (17.8% of all deaths in India) attributable to air pollution in 2019.
- Outdoor particulate pollution death rate up 115% from 1990 to 2019.
- Economic loss US$36.8 billion (1.36% of GDP); Telangana about US$1.1 billion, US$28.7 per person.

How we use it: The health and money cost of the problem, including for our state.

05. Design of CANSAT for Air Quality Monitoring for an altitude of 900 meters

Soma Kunal Raj, Yalamanchili Surya Teja (2024). arXiv:2409.03727. Vidya Jyothi Institute of Technology, Hyderabad.

What they did: Built a 310 x 125 mm, about 1 kg CanSat (NAMBI-VJ) with a mechanical gyroscope and spill-hole parachute, dropped from 900 m by drone, sending data over Zigbee.

Key findings:

- Measured gas levels (PPM, CO2) and GPS; descent rate 3 m/s.
- Shows a Hyderabad team has flown an air quality CanSat from a drone.
- No PM2.5 sensor and no comparison with a reference station.

How we use it: Prior art we build on; our sensor validation and 700 g design are the step beyond it.

06. Design and Implementation of an Efficient Onboard Computer System for CanSat Atmosphere Monitoring

Gadekar A. (2023). arXiv:2308.03496. MIT World Peace University, Pune.

What they did: Designed a CanSat onboard computer around an ESP32 with GPS, motion, pressure, temperature, UV and gas sensors and a transceiver.

Key findings:

- Uses an ESP32 for low power and built-in connectivity.
- Focus is the electronics design, not a measured PM2.5 profile.

How we use it: Supports our choice of an ESP32 as the CanSat's computer.

07. Air Quality Life Index: India Fact Sheet 2025

Energy Policy Institute at the University of Chicago (EPIC), 2025. Data year 2023.

What they did: Satellite-based PM2.5 estimates turned into life expectancy lost, for India and each state.

Key findings:

- India's PM2.5 in 2023: 41 µg/m³; meeting WHO's guideline would add 3.5 years to life expectancy.
- All 1.4 billion Indians live above the WHO guideline; 46% above the national standard.
- Telangana: 29.3 µg/m³, 2.4 years of life lost relative to WHO.

How we use it: Headline national numbers on slides 5 and 6.

08. Tracing the Hazy Air 2026: Progress Report on National Clean Air Programme (NCAP)

Centre for Research on Energy and Clean Air (CREA), January 2026.

What they did: Tracks NCAP progress, India's monitoring network, and city PM10 and PM2.5 levels for 2025.

Key findings:

- 1,600 stations in 2025 (565 continuous, 1,035 manual); manual target short by 465.
- 28 of 130 NCAP cities have no continuous station; data quality is a concern.
- 103 of 231 cities above the PM2.5 standard; all monitored Telangana cities above the PM10 standard.

How we use it: Evidence of gaps in the ground network.

09. WHO Global Air Quality Guidelines (2021)

World Health Organization, 2021. ISBN 9789240034228.

What they did: The global guideline values for PM2.5, PM10, ozone, NO2, SO2 and CO, with interim targets.

Key findings:

- PM2.5 annual guideline 5 µg/m³; interim targets 35, 25, 15, 10.
- PM2.5 24-hour guideline 15 µg/m³; interim targets 75, 50, 37.5, 25.

How we use it: The yardstick in every chart.

10. Design your parachute: a guide to landing your CanSat safely (Teach with Space T10)

European Space Agency (ESA) Education, 2023.

What they did: Teaching guide for sizing a CanSat parachute from the forces at terminal speed.

Key findings:

- Terminal speed balance: $m \cdot g = \frac{1}{2} \cdot Cd \cdot \rho \cdot A \cdot v^2$.
- Drag coefficients: hemispherical 0.62, cross 0.8, flat hexagon 0.8.
- ESA's own competition asks for 8 to 11 m/s; our problem statement asks for 5 to 8 m/s.

How we use it: How we size our parachute (chart D3).

Part G. Data and method

Where the Hyderabad data comes from

- CPCB and Telangana State Pollution Control Board continuous stations, as published by OpenCity (data.opencity.in, dataset 'Hyderabad Hourly Air Quality Reports', 15-minute files for 2024-25).
- 14 stations: Bollaram Industrial Area, Central University, ECIL Kapra, ICRISAT Patancheru, IDA Pashamylaram, IITH Kandi, Kokapet, Kompally Municipal Office, Nacharam TSIIC IALA, New Malakpet, Ramachandrapuram, Sanathnagar, Somajiguda, Zoo Park.
- 70,176 time slots per station (1 Jan 2024 to 31 Dec 2025, every 15 minutes).

What we did

- Kept PM2.5 values between 0 and 1,000 µg/m³; counted the share present (completeness).
- Daily average only for days with at least 48 of 96 valid readings (half the day).
- Yearly average = mean of valid daily averages. Days over a limit counted on valid days only.
- Hour-of-day curves for winter (Nov to Feb) and monsoon (Jun to Sep), using the station's own timestamps (local time).
- City series = median across the 6 stations with a normal seasonal pattern, for days with 3 or more of them reporting. A median limits the effect of spikes such as Sanathnagar's.
- Correlation (r) between daily PM2.5 and daily relative humidity at each station.

Limits of this analysis

- We did not correct or remove any station's data. Flags in Part B are signs to check, not proof of a fault.
- Station inlets sit a few metres above ground; it says nothing directly about air higher up. That is the gap our CanSat targets.
- This is two years of data; one year can differ a lot from the next (Zoo Park changed by 18 µg/m³).

Files in the data folder

File	Contents
hyderabad_station_summary_2024_25.csv	One row per station: yearly averages, days over limits, completeness, winter-to-monsoon ratio, highest day, humidity correlation
hyderabad_winter_vs_monsoon_hourly.csv	Hour-of-day PM2.5 averages for winter and monsoon, per station
hyderabad_daily_median_6_stations.csv	Daily PM2.5 median of the 6 seasonal stations, 2024-25

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